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Project Reflection

To conclude the module, we demonstrated our project and gave a presentation. Leading up to the project we learned many important skills, such as using microcontrollers, programming, proper use of AI, prototyping, GitHub documentation, and soft skills such as team work. All of these skills contributed to the success of our project and the delivery of our presentation.

Our goal of creating an automated shower vent system was successful. We met all of our established requirements and stayed within the constraints for the project. Factors that contributed to the success of our project were division of responsibilities and good communication. When beginning the project, we established roles for each team member. This was ultimately highly effective, as it allowed us to get work done much faster while maintaining communication across each role.

Some difficulties we encountered were issues interfacing electronics with our provided materials and unreliability of our PEDOT sensor. We originally intended to use the DC motor to power the vent fan, but after lots of trial and error we found that it was not possible with our available materials. The Arduino Nano ESP32 was not able to supply enough power for the motor on any logic pins, so the only way to power it was through VBUS with no way of turning the motor on or off. Due to this, we switched to using a stepper motor. This required us to rewrite and debug our program, which we did using Claude. We used the PEDOT sensor to detect changes in humidity by detecting raw ADC values and comparing this to a chosen threshold. However, the readings from the sensor were very volatile, likely due to poor electrical contact with the diode

or noise interference. To account for this, we included a push-button switch to manually engage the system. Having solved these problems, we were able to successfully demonstrate our project. Lastly, we successfully delivered our presentation to the class.